Zachary Leong

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Education

University of Pennsylvania — Philadelphia, PA

B.S.E. in Computer Science, major in Digital Media Design (GPA: 3.96/4.00) — *May 2028*

- Coursework: Data Structures and Algorithms, Operating Systems, Computer Systems, Linear Algebra, 3D Modeling, Computer Animation, Procedural Design Systems

M.S.E. in Computer Graphics & Game Technology — *May 2028*

- Coursework: Offline & Real-time Rendering

Experience

Aviation Systems Software Engineering Intern — June 2026 – August 2026

NASA Ames Research Center — *Mountain View, CA*

- Prototyped a Python modernization of *Freddie*, a drone traffic management and airspace automation system
- Designed 4 microservices with Kafka messaging, REST APIs, WebSocket updates, and Postgres storage
- Built real-time drone telemetry simulation, conformance checking, and spatial visualizations for flight paths and 4D airspace volumes

Teaching Assistant — January 2025 – Present

Computer Graphics & Discrete Mathematics — *Philadelphia, PA*

- Help students debug C++ graphics projects: scene graph, rasterization, shaders, OpenGL, half-edge mesh
- Previously led weekly discrete math recitations of 15 students in logic, graph theory, and probability

Research Intern — October 2024 – December 2025

GRASP Sung Robotics Lab — *Philadelphia, PA*

- Independent Study: *GPU-Accelerated 3D CSC Dubins Path Distance Checking using Signed Distance Fields*
  - Derived and implemented SDFs for CSC paths to enable differentiable collision costs for gradient descent
  - Benchmarked scaling across 50+ links and 10M+ points, validating GPU scalability over CPU methods
- Implemented custom 3D transform gizmos for precise local/world-space manipulation of kinematic linkages using PyQtGraph, supporting a research paper submitted to *IROS 2026*

Projects

Space Minecraft — *C++, GLSL, OpenGL, Qt*

- Collaborated in a team of three to create a voxel game engine supporting spherical planets in a solar system
- Contributed debug GUI, shadow maps, biome blending, procedural planets, and foliage scattering
- Optimized performance by using compute shaders, multithreading, terrain chunking, and console profiling

GPU Monte Carlo Path Tracer — *C++, GLSL, OpenGL, Qt*

- Engineered a GLSL unidirectional path tracer with BVH acceleration (SAH) and thin-lens depth of field
- Implemented microfacet BRDF, Multiple Importance Sampling, and Next Event Estimation
- Integrated OpenImageDenoise and QMC low-discrepancy sampling, improving convergence times by 2x

Evolved Creature Simulation — *Python, PyBullet*

- Designed genetic algorithms to evolve and breed articulated creature morphologies and neural controllers

Additional Links: Spring 2026 Demo Reel, Complex Origami Showcase

Technical Skills

Languages: C++, Rust, Python, Java, HTML, CSS, JavaScript, C, C#, OCaml, VEX, HLSL, GLSL
Platforms and Tools: OpenGL, WebGPU, Nsight Graphics, RenderDoc, Qt, Linux, Git, Claude Code, CUDA
Software: Houdini, Blender, Maya, Substance Painter, DaVinci Resolve, Photoshop, Figma, Unity, Unreal Engine